Constructor University

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Lab Experiment 6: Operational Amplifiers

Author of the report: MR. Giorgi Kipshidze

Experiment conducted by	: Giorgi Kipshidze, Jung Woojin
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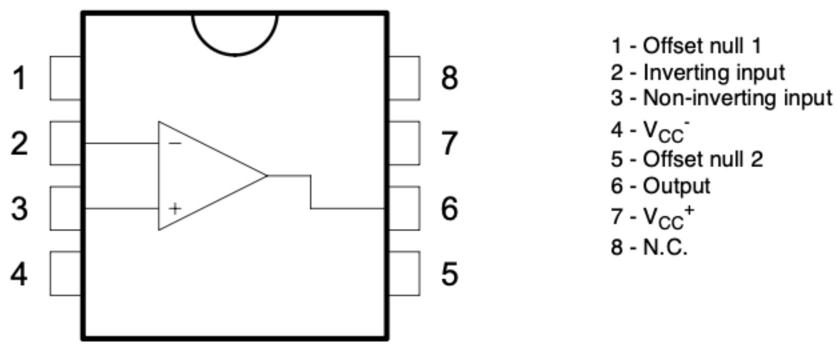
1. Introduction

The OpAmps are high-gain voltage amplifiers used in analog circuits. In the ideal model, the input resistance is infinite, the output resistance is zero, and the open-loop gain is very large. The OP-Amp is an active component and needs a supply-voltage to operate with negative feedback, an op-amp can be made to work as an amplifier with a predictable gain, a voltage follower, an integrator, a summing circuit, or an active filter.

In this experiment the UA741 operational amplifier was used. Measurements were made with a function generator and a two-channel oscilloscope. The input was observed on channel 1 and the output on channel 2.

UA741 OpAmp Schematics.



In this experiment two circuits were investigated. The first circuit was an inverting amplifier. For an ideal inverting amplifier, the voltage gain is determined by the ratio of the feedback resistor to the input resistor: $A_{gain} = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$ negative values means it is inverted.

And the second circuit was an Op-Amp integrator. In this circuit, a capacitor is placed in the feedback path, so the output voltage is related to the integral of the input voltage. The theoretical peak-to-peak output voltage is given by: $V_{outpp} = \frac{V_{in(pp)}}{4fR_2C_1}$

2. Execution

2.1 Experiment setup

Workbench No.6

Used tools and instruments:

- Breadboard
- UA741 operational amplifier
- DC power supply

- Function generator
- Oscilloscope
- Resistors: 50Ω, 1kΩ, 10kΩ, 22kΩ, 100kΩ
- Capacitor: 100nF

The UA741 operational amplifier required a positive and negative supply voltage. Therefore, the general supply circuit was first assembled using +10V, -10V, and two 100nF capacitors.

For the oscilloscope measurements, channel 1 was connected to the input signal and channel 2 was connected to the output of the operational amplifier. The phase shift was measured between channel 1 and channel 2. The manual also instructed to use AC coupling, the oscilloscope averaging function to reduce noise, and the measurement function to record amplitudes and phase shifts faster.

2.2 part 1: Inverting Amplifier-Setup

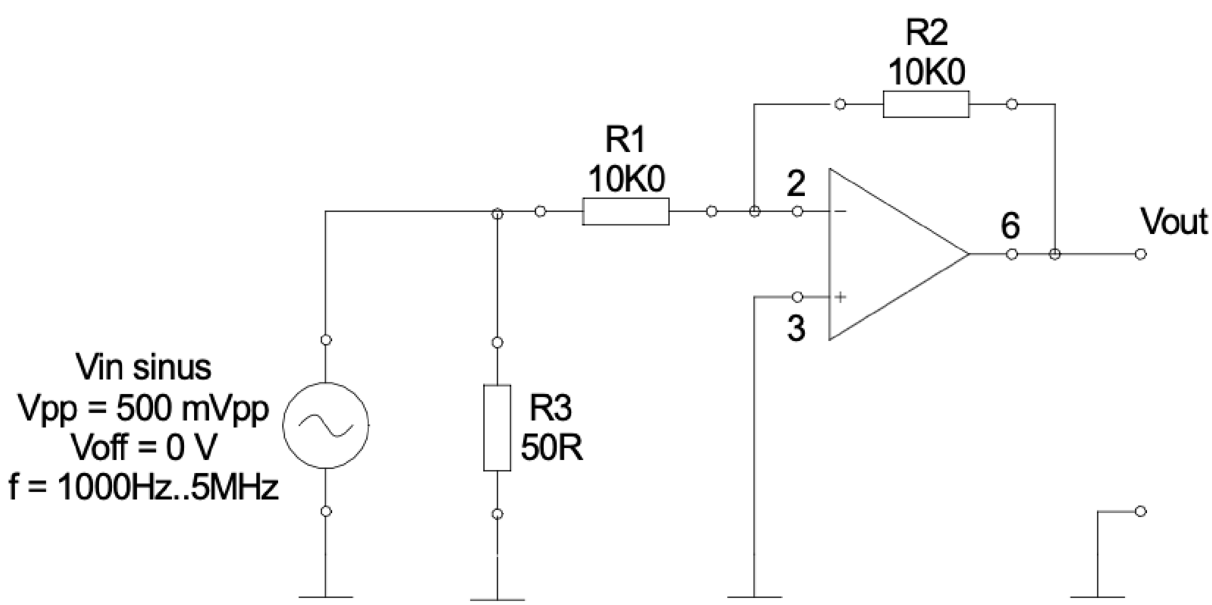
In the first experiment we used operational amplifier as inverting amplifier.

The signal generator's settings where:

$V_{in} = 500mV_{pp,}V_{off} = 0V, and\ frequecy\ was\ varied\ from\ f = 1000Hz\ to\ 5mHZ$

The feedback resistor was $R_2 = 10k\Omega$.

The input resistor was first $\mathrm{R}_1 = 10\mathrm{k}\Omega$ and then it was with $\mathrm{R}_1 = 22k\Omega$ and then $R_1 = 1k\Omega$



The channel 1 of the oscilloscope was connected to the input and the channel 2 connected to output of the OpAmp.

Theoretical gains were calculated: $A_{v1} = -\frac{10k\Omega}{10k\Omega} = -1$ $A_{v2} = -\frac{10k\Omega}{22k\Omega}$=-0.45 $A_{v3} = -\frac{10k\Omega}{1k\Omega}$=-10

2.3 Part 1 : Inverting Amplifier-Execution and Results

The frequency was increased form 1kHZ to 5MHz. At each frequency, the input amplitude, output amplitude, and phase shift between channel 1 and channel 2 were recorded.

Inverting Amplifier with: $R_1 = 10k\Omega,\ R_2 = 10k\Omega$

Part 1 : Inverting Amplifier			
Resistor = 10k			
Frequency (Hz)	Input Amplitude (mV)	Output Amplitude (mV)	Phase (degree) *reference ch2-ch1
1k	248	256	-179
2k	248	256	-178
5k	248	256	-179
10k	248	256	-179
20k	248	256	-177
50k	248	256	-168
100k	248	248	-156
200k	248	232	-126
500k	248	114	-68.4
1M	248	51.2	-1.8
2M	248	15.2	22.3
5M	248	12	5.03

At low frequency, the output voltage was almost equal to the input voltage. This agrees with the theoretical gain of -1. The phase shift was also close to 180, meaning that the output signal was inverted.

Inverting Amplifier with: $R_1 = 22k\Omega,\ R_2 = 10k\Omega$

Resistor = 22k			
Frequency (Hz)	Input Amplitude (mV)	Output Amplitude (mV)	Phase (degree) *reference ch2-ch1
1k	248	116	179
2k	248	116	-178
5k	248	120	-177
10k	248	120	-174

20k	248	120	179
50k	248	120	-176
100k	248	120	-169
200k	248	124	-155
500k	248	98	-130
1M	248	76	-58.2
2M	248	36.8	-36
5M	232	152	-45.2

For this resistor combination, the theoretical magnitude of the gain is 0.455. At low frequency, the measured output amplitude was around 116–120,mV, while the input amplitude was 248mV. This gives a measured gain close to the theoretical value.

Inverting amplifier with: $R_1 = 1k\Omega,\ R_2 = 10k\Omega$

Frequency (Hz)	Input Amplitude (V)	Output Amplitude (V)	Phase (degree) *reference ch2-ch1
1k	240mV	2.4V	-177
2k	248mV	2.48V	-179
5k	248mV	2.4V	-177
10k	248mV	2.4V	-170
20k	248mV	2.32V	-164
50k	248mV	1.8V	-131
100k	248mV	1.04V	-108
200k	248mV	540mV	-90
500k	256mV	216mV	-77
1M	256mV	186mV	-54.8
2M	252mV	69.6mV	-24.4
5M	224mV	-424mV	-47.4

For R_1 = 1kΩ, the expected gain magnitude was 10. At low frequencies, the output was around 2.4V for an input of about 240--248,mV, so the measured gain was close to 10. However, the gain decreased strongly as the frequency increased. This shows the practical bandwidth limitation

2.4 Part 2 : The Op-Amp Integrator-Setup

In the second part, the operational amplifier was used as an inverting integrator. The feedback path contained a capacitor, so the output voltage depended on the integral of the input signal. A square-wave input was used, and the output waveform was observed on the oscilloscope.

Used components were: $R_1 = 100k\Omega\ R_2 = 100k\Omega\ \ C_1 100nF\ and\ OpAmps\ supply\ was\ set\ \pm 10V$

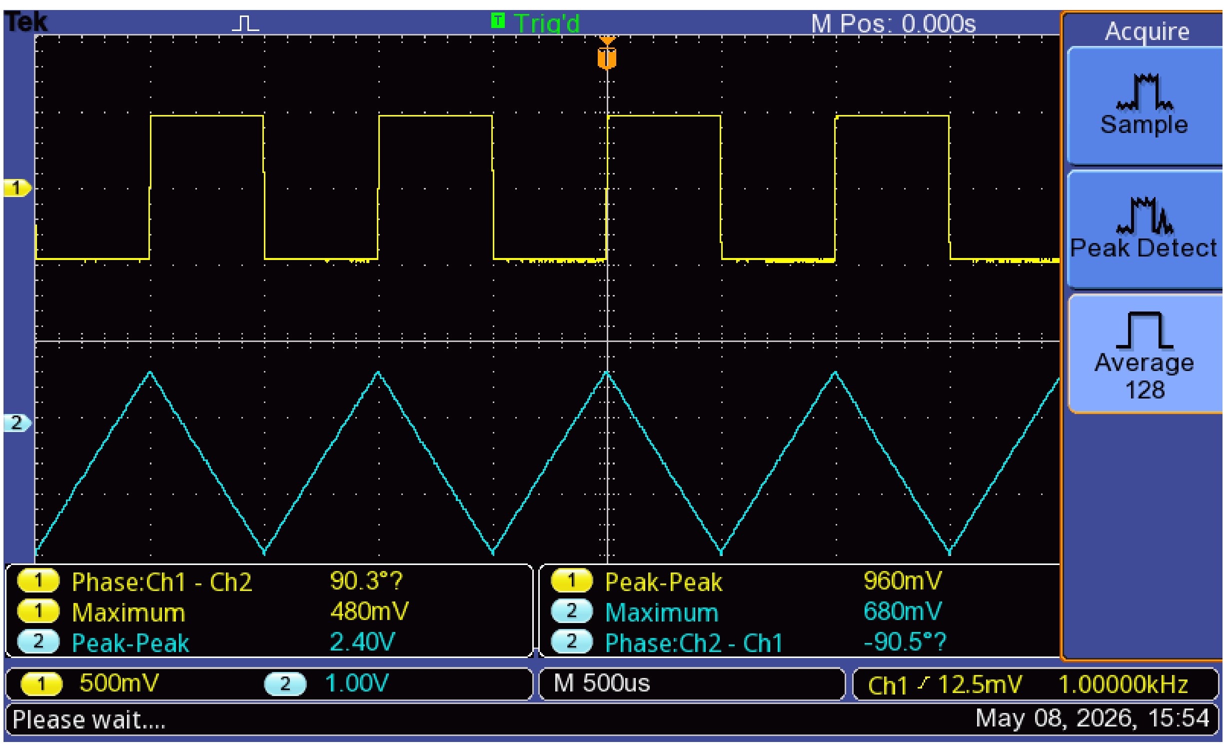
Ch1 of the oscilloscope was used for the input signal, and Ch2 was used for the output signal.

The signal generator was set to: Square wave: $V_{in} = 1V_{pp}\ and\ V_{off} = 0V$

The frequency was changed from 1kHz to 5kHz. The goal was measuring the input and output amplitudes at 1kHz, describing the shape of the output waveform, saving an oscilloscope screenshot, then repeating the output measurement and screenshot at 5kHz.

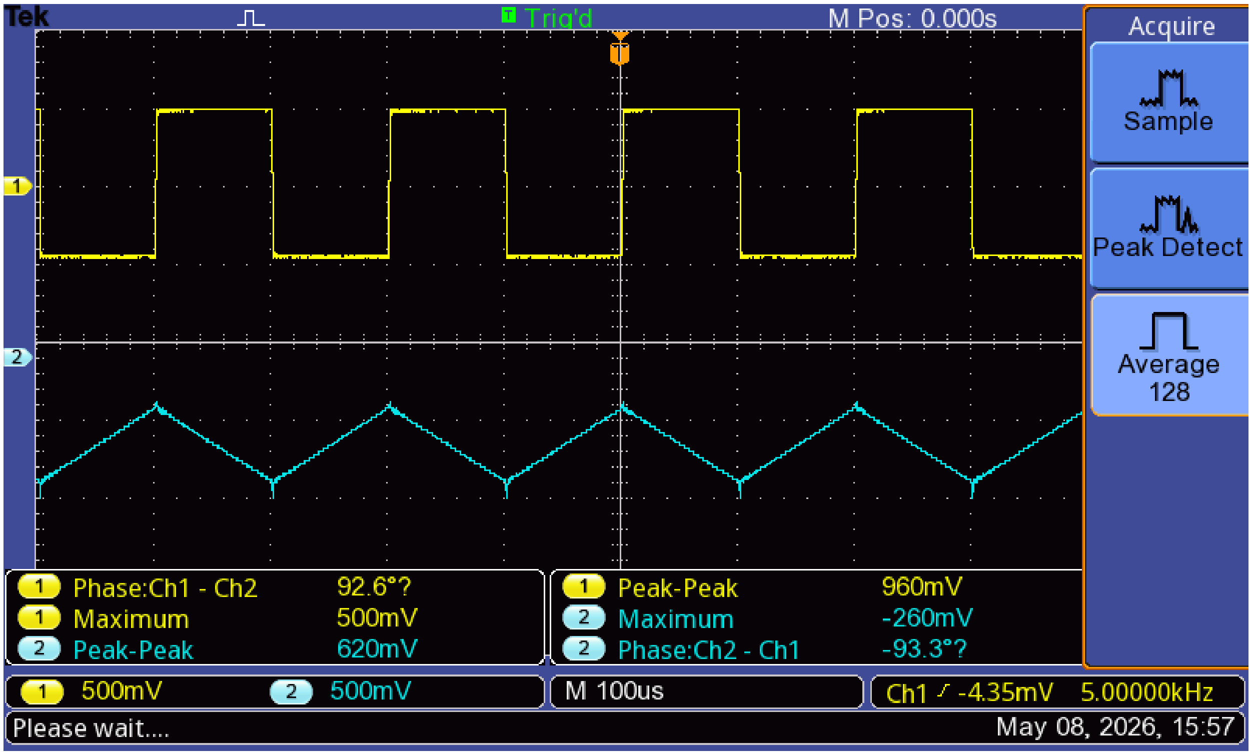
2.5 Part2 : The OpAmp Integrator – Execution and Result

At 1kHz, the input and output waveforms were measured on the oscilloscope. The input waveform was a square wave and the output waveform was a triangular wave.



This agrees with the behavior of an integrator, because the integral of a constant positive or negative voltage is a linear ramp.

At 5kHz, the same measurement was repeated. The output waveform remained triangular, but the output amplitude became smaller.



Integrator Measurements

Frequency	Input Waveform	Output waveform	Measured Vin(pp)	Measured Vout(pp)
1kHz	Square wave	Triangular wave	480mV	680mV
5kHz	Square wave	Triangular wave	540mV	280mV

The output waveform was triangular, as expected for an integrator with a square-wave input. The measured output amplitude decreased when the frequency increased from 1kHz to 5kHz. This agrees with the theoretical relationship

3. Evaluation

3.1 Part 1 : Inverting Amplifier - Evaluation

For the inverting amplifier, the theoretical voltage gain is: $A_v = -\frac{R_2}{R_1}$

For the first resistor Combination: $R_1 = 10k\Omega$ and $\mathrm{R_2} = 10k\Omega$ $\mathrm{A_v} = -\frac{10k\Omega}{10k\Omega} = -1$

The measured low-frequency gain was $|A| = \frac{0.500}{0.504} = 0.992$

The percentage error is: 0.8%

For the second resistor combination is: $R_1 = 22k\Omega\ and\ R_2 = 10k\Omega$ $A_v = -\frac{10k\Omega}{22k\Omega} = -0.455$

The measured low-frequency gain was: $|A| = \frac{0.228}{0.504} = 0.452$

The percentage error is: 0.66%

For the third resistor combination: $R_1 = 1k\Omega\ and\ R_2 = 10k\Omega$ $\mathrm{A_v} = \frac{4.88}{0.488} = 10$

The measured values are summarized :

R_1	R_2	Theoretical |A|	Measured |A|	Error
10k	10k	1	0.992	0.80%
22k	10k	0.455	0.452	0.66%
1k	10k	10	10	0.00%

From the table, the low-frequency results match the theoretical gain formula very well. In all three cases, the phase shift was close to 180, so the circuit behaved as an inverting amplifier.

At higher frequencies, the gain decreased. This was especially visible for the gain of 10 circuit. For example, at 100kHz: : $|A| = \frac{1.84}{0.504} = 3.65$

This is much smaller than the low-frequency value of 10. This happens because the UA741 is not an ideal Op-Amp. It has limited bandwidth, so as frequency increases, the available gain decreases.

The phase shift also changed strongly at high frequencies. Instead of staying close to 180, it moved toward smaller values. This confirms that the real Op-Amp has frequency limitations.

3.2 part 2 : Op-Amp Integrator - Evaluation

$V_{out} = \frac{V_{in(pp)}}{4fR_2C_1}$ Component values used were: $R_2 = 10000\Omega$. $C_1 = 100nF$ $V_{in} = 1V$

For 1kHz: $V_{out(pp)} = \frac{1}{4\times1000\times10000\times10^{-10}} = \frac{1}{4} = 0.25V$

For 1kHz: $V_{out(pp)} = \frac{1}{4\times5000\times10000\times10^{-10}} = \frac{1}{20} = 0.05V$

The theoretical results are:

Frequency	Theoretical V_{out(pp)}
1 kHz	0.25 V
5 kHz	0.05 V

The measured oscilloscope values were larger than the theoretical values, but the trend was correct. The output voltage decreased when the frequency increased. At 1kHz, the output was much larger than at 5kHz, which agrees with the inverse proportionality:

$$V_{out} \sim \frac{1}{f}$$

The output waveform was triangular because the input was a square wave. During the positive part of the square wave, the capacitor current is approximately constant, so the capacitor voltage changes linearly. During the negative part of the square wave, the current changes direction, so the output ramp changes direction. This creates a triangular output waveform.

The manual asks why the output ramps up when the input goes negative and ramps down when the input goes positive.   This happens because the circuit is an inverting integrator. A positive input produces a negative slope at the output, and a negative input produces a positive slope.

The manual also asks about the 100k\Omega resistor connected in parallel with the capacitor.   This resistor provides a DC feedback path. Without this resistor, small input bias currents and input offset voltage of the UA741 would charge the capacitor over time. The output would drift and eventually saturate near the supply voltage. With the resistor included, the capacitor can discharge slowly, and the operating point stays stable.

4. Conculusion

In this experiment, the UA741 operational amplifier was used in two circuits from the manual: an inverting amplifier and an Op-Amp integrator.

In the first part, the inverting amplifier was tested with three resistor ratios. The measured low-frequency gains matched the theoretical formula: $A_v = -\frac{R_2}{R_1}$

For R_1=10kΩ, the gain was approximately 1. For R_1=22kΩ, the gain was approximately 0.455. For R_1=1kΩ, the gain was approximately 10. The phase shift was close to 180, confirming the inverting behavior.

At higher frequencies, the gain decreased and the phase shifted away from 180^\circ. This shows the practical limitations of the UA741, especially its finite bandwidth. The gain of 10 circuit lost gain earlier than the lower-gain circuits, which agrees with the gain-bandwidth limitation of real Op-Amps.

In the second part, the Op-Amp integrator was tested using a square-wave input. The output waveform was triangular, as expected, because the integral of a square wave is a ramp. The output amplitude decreased when the frequency increased from 1kHz to 5kHz, which agrees with the manual equation: $V_{out} = \frac{V_{in(pp)}}{4fR_2C_1}$

The measured values were not exactly the same as the theoretical values. The differences may come from resistor and capacitor tolerances, probe, oscilloscope measurement uncertainty, non-ideal UA741 behavior, and the exact input amplitude. Overall, the experiment confirmed the basic behavior of Op-Amp feedback circuits and showed the difference between ideal and real operational amplifiers.

5. Reference

Uwe, P. Electrical Engineering II Lab Manual, CH-211-B, Constructor University Bremen, Spring Semester 2026.